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SUPERSEDING
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3 June 1952

MILITARY SPECIFICATION

MUSTARD, H (LEVINSTEIN PROCESS)

This specification has been approved by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 This specification covers one grade of bis(2-chloroethyl) sulfide $(ClCH_2CH_2)_2S$.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-R-191 —Red-Lead; Dry and Paste-in-Oil.
TT-W-261 —White - Lead; Basic - Sulphate, Dry and Paste-in-Oil.

MILITARY

JAN-H-502—Hexamine (For Use in Ammunition).
MIL-C-3250—Container, One-Ton, Type D.

U. S. ARMY

96-31-64 —Container, One-Ton, Type E.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129—Marking of Shipments.

DRAWINGS

CHEMICAL CORPS

E6-2-2—Drum, 55-Gallon, Type B, Assembly and Bill of Material.

283307—52

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.**—The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

INTERSTATE COMMERCE COMMISSION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Chemical and physical.**—The material shall conform to table I.

TABLE I.—*Chemical and physical requirements*

Requirement	Maximum (percent)	Minimum (percent)
Iron (as $FeCl_3$), by weight	0.25	
Acidity (as HCl), by weight	0.30	
H content, by weight		68.0
Distillation loss, by weight	3.0	

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.**

4.1.1 *Batch process.*—A lot shall consist of the material produced from no more than one manufacturing batch (6.3).

4.1.2 *Continuous process.*—A lot shall consist of the material produced in no more than one work shift with no change in process or materials.

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4.2 **Sampling.**—Five, 1-pound specimens shall be selected as representative of each lot. The specimens may be taken randomly from containers or during the container filling operation. The specimens shall be placed in clean, dry containers, and labeled to identify the container and lot represented. Each specimen shall be tested as specified in 4.4 (see 6.4, 6.5, and 6.6).

4.3 **Inspection.**

4.3.1 *Packing and marking.*—The inspector shall inspect the packing and marking for compliance with section 5.

4.3.2 *Procedures.*—For Navy purchases, the general inspection procedures shall be in accordance with the General Specifications for Inspection of Material.

4.4 **Tests.**

4.4.1 *Reagents.*—Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the test, and corrections shall be applied when significant.

4.4.2 *Ferrous chloride.*—Introduce about 10 ml. of the specimen in a dry, tared, 600-ml. beaker, provided with a watch-glass cover, and weigh the container and contents to the nearest mg. Cautiously add approximately 20 ml. of concentrated nitric acid, dropwise from a pipette, agitating frequently, until the violent frothing subsides. Add 20 ml. more of the concentrated nitric acid, and heat until the frothing ceases and the liquid is pale yellow. Cool slightly, add 15 ml. of 70-percent perchloric acid, and heat with frequent agitation until dense fumes of the perchloric acid are given off and the solution is light yellow. Cool and dilute to approximately 150 ml. with water. Add an excess of 6*N* ammonium hydroxide and boil for 5 minutes. Filter while warm and wash the precipitate five times with a hot 2-percent ammonium chloride solution and discard the filtrate. Dissolve the precipitate by pouring 30 ml. of hot 3*N* hydrochloric acid through the filter, collecting the filtrate in a 250-ml. beaker or casserole. Using the Zimmermann-Reinhardt method, reduce with stannous chloride and titrate with 0.1*N* potassium permanganate solution. Calculate the percent iron as $FeCl_3$ as follows:

$$\text{Percent } FeCl_3 = \frac{12.7 \times A \times B}{C}$$

where: A = ml. of potassium permanganate solution
B = normality of potassium permanganate solution
C = weight of specimen in gm.

4.4.3 *Acidity.*—Place approximately 5 ml. of the specimen in a dry, tared, 250-ml. iodine flask, and reweigh the flask to the nearest mg. Wash down the stopper and sides of the flask with approximately 15 ml. of chloroform and add approximately 50 ml. of a saturated sodium chloride solution. Stopper the flask, shake well, add 10 drops of phenolphthalein indicator, and titrate with 0.1*N* sodium hydroxide solution. Do not shake while titrating, but roll or rock the flask gently to mix the contents. The pink color, upon gentle rotation of the flask, shall persist throughout the upper layer for 10 seconds before an end point is declared. Run a blank on the same quantities of chloroform and sodium chloride solutions, and substract the blank from the titer. Calculate the acidity as follows:

$$\text{Percent acidity as HCl} = \frac{(A-B) \times C \times 3.65}{D} - (0.576 \times E)$$

where: A = ml. of sodium hydroxide solution required for specimen
B = ml. of sodium hydroxide solution required for blank
C = normality of sodium hydroxide solution
D = weight of specimen in gm.
E = percent ferrous chloride (4.4.2).

4.4.4 *Purity.*

4.4.4.1 *Vacuum distillation.*—Place approximately 95 gm. of the specimen, weighed to the nearest dg., in a tared, Claissen flask. Assemble the vacuum-distillation apparatus as shown on figure 1, and test for leaks at a pressure of 5±1 mm. of mercury. Conduct the distillation at a pressure of 5±1 mm. of mercury, allowing a small amount of air to pass through the system. Apply heat gradually to the oil bath, and distill the mustard at a rate of 60 drops per minute until the bath reaches a temperature of 175° C.

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Hold the bath at this temperature until the rate of distillation falls to 1 drop per minute; then discontinue the distillation. Weigh the distillate and residue, and calculate the percent distillate, residue, and loss. Reserve the distillate in the side-arm test tube for the freezing-point determination (4.4.4.2).

4.4.4.2 *Freezing point.*

4.4.4.2.1 *Apparatus.*—The apparatus for determining the freezing point is shown on figure 2. The thermometer used should previously have been checked against a standardized thermometer for corrections in the appropriate temperature range.

4.4.4.2.2 *Procedure.*—Fill the 1⅜-inch test tube with the specimen to the immersion mark on the thermometer. If an immersion thermometer is not used, fill to ½ inch above the thermometer bulb. In no case should less than 25 ml. be used. Assemble the thermometer and stirrer in the cork stopper and place in the test tube. The temperature range on the thermometer to be observed should be above the cork. Place the test tube in the outer jacket and immerse in a bath at approximately 5° C. below the anticipated freezing point. Allow the temperature of the mustard to fall to 1.0±0.1° C. below the estimated freezing point of the material while stirring. If crystallization does not occur before this point, it must be induced by vigorous stirring or by seeding with previously frozen crystals of the same material. Record the highest temperature attained during crystallization as the freezing point. If a total immersion thermometer is used, correct for emergent stem. Determine the purity of the mustard on which the freezing point was obtained by the following formula, in which it is assumed that for each percent of impurity in the distillate, the freezing point is lowered 0.368° C. The freezing point of pure mustard is taken as 14.45° C.

$$\text{Percent purity of distillate} = 100 - \frac{(14.45 - A)}{0.368}$$

where: A = freezing point of specimen in ° C.

4.4.4.3 *Calculation.*—Calculate the purity of the specimen as follows:

$$\text{Percent purity of specimen} = \frac{A \times B}{C}$$

where: A = weight of distillate, in gm.
B = percent purity of distillate (4.4.4.2.2).
C = weight of specimen in gm. (see 4.4.4.1).

4.5 **Rejection and resubmission.**—If any specimen fails to conform to this specification, the lot shall be rejected. The contractor, at no expense to the Government shall have the option of having an analysis made on each container in the lot, removing the nonconforming material, and resubmitting the remaining portion of the lot for acceptance testing. If any specimen taken from the resubmitted portion of the lot fails to conform to this specification, the lot shall be finally rejected.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**

5.1.1 *One-ton container.*—The mustard shall be packed in 1-ton containers conforming to Specification 96–31–64 or MIL–C–3250. Before filling, the container shall be examined for the presence of foreign material, cleaned, and dried if necessary. Each 1-ton container shall contain 170±3 gallons of the material. All threads of fittings, and threaded openings in the container shall be cleaned, and the male threads shall be luted with basic-sulphate, white-lead, paste-in-oil conforming to Specification TT–W–261 or red-lead, paste-in-oil conforming to Specification TT–R–191.

5.1.2 *Fifty-five gallon drums.*—When the mustard is packed for immediate use, or for transfer to a processing plant, it may be packed in 55-gallon steel drums conforming to Drawing E6–2–2. The drums shall contain 50±1 gallons of the material.

5.1.3 *Stabilization.*—If the mustard is being packaged for storage (not for use in a thickening, filling, purification, or other processing plant) it shall be stabilized by the addition of 1.0±0.1 percent by weight of hexamethylenetetramine conforming to Specification JAN–H–502. The stabilizer may be added either before or after (preferably after) the mustard has been placed in the container.

5.2 **Marking.**—In addition to any special marking required by the contract or order, all shipping containers shall be marked in accord-

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ance with Regulations for Transportation of Explosives and Other Dangerous Articles, etc., and Standard MIL-STD-129. If the mustard in a 1-ton container, or a 55-gallon drum, has been stabilized as specified in 5.1.3, the container or drum shall be marked with the symbol "+X" in letters 1 inch high on the valve end of the 1-ton container or the bung end of the 55-gallon drum

6. NOTES

6.1 Intended use.—The mustard covered by this specification is intended for use as a chemical agent.

6.2 Ordering data.—Procurement documents should specify:

(a) Title, number, and date of this specification.

(b) Type of packing required (see 5.1).

6.3 Batch.—A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 Sampling and testing.—When the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.5 Changes in composition.—Levinstein mustard, in storage, is subject to changes in composition, in addition to deposition of sulfur. If for any reason the contents of a container are analyzed, care should be taken that a representative sample is obtained. · If the mustard has frozen and then thawed, the contents of the container tend to stratify, and vigorous agitation is required to obtain uniform material.

6.6 Precautions in handling.—Mustard in the liquid or vapor phase is poisonous and vesicant and must be handled with extreme care. A safety directive covering the safe handling of mustard, incident to sampling or testing, will be furnished the contractor upon request.

6.7 Safety classification.—This item carries Ordnance safety classification 11, storage group A.

6.8 Stock number.—This item carries Chemical Corps stock number 324110.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Chemical Corps
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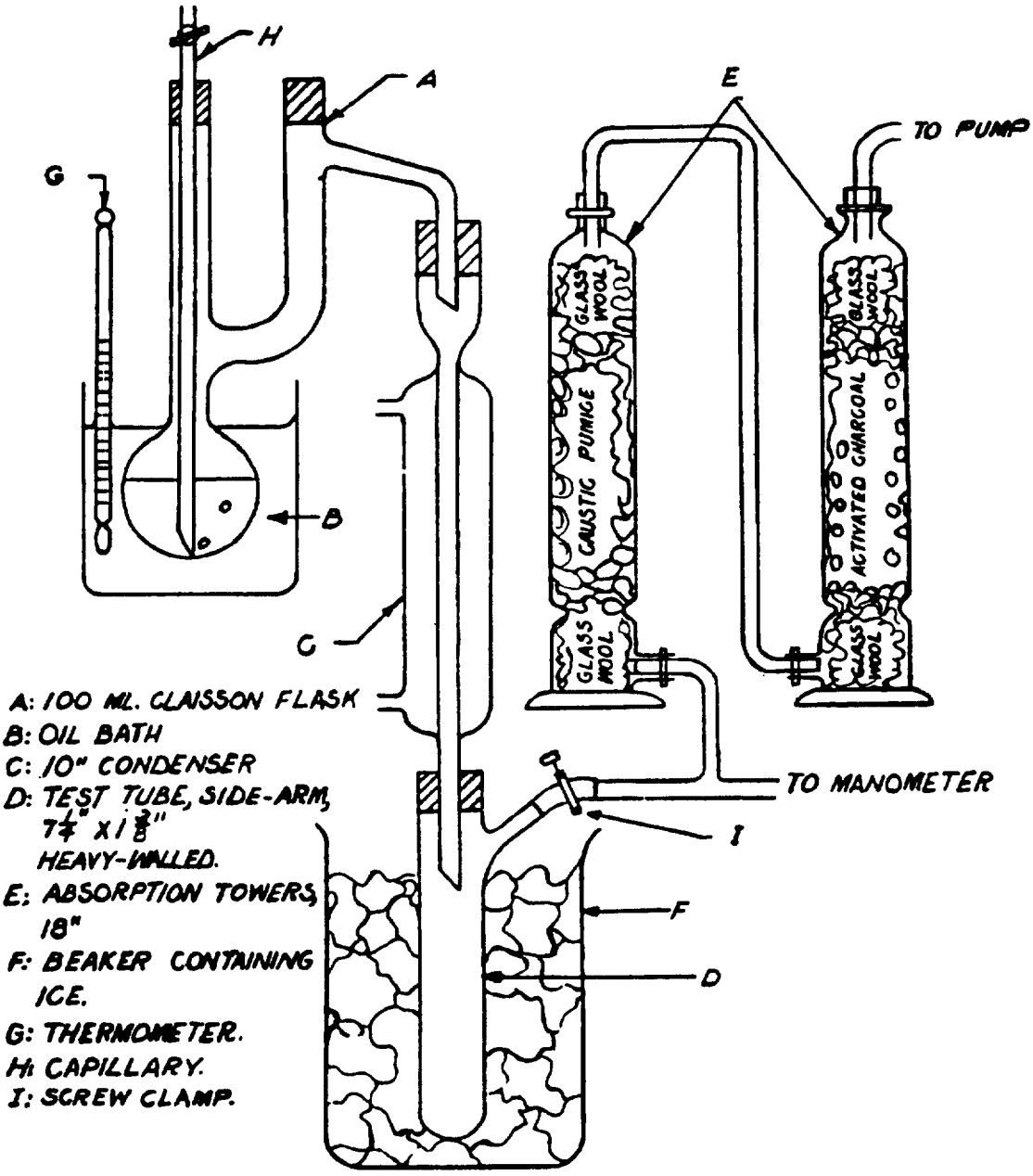
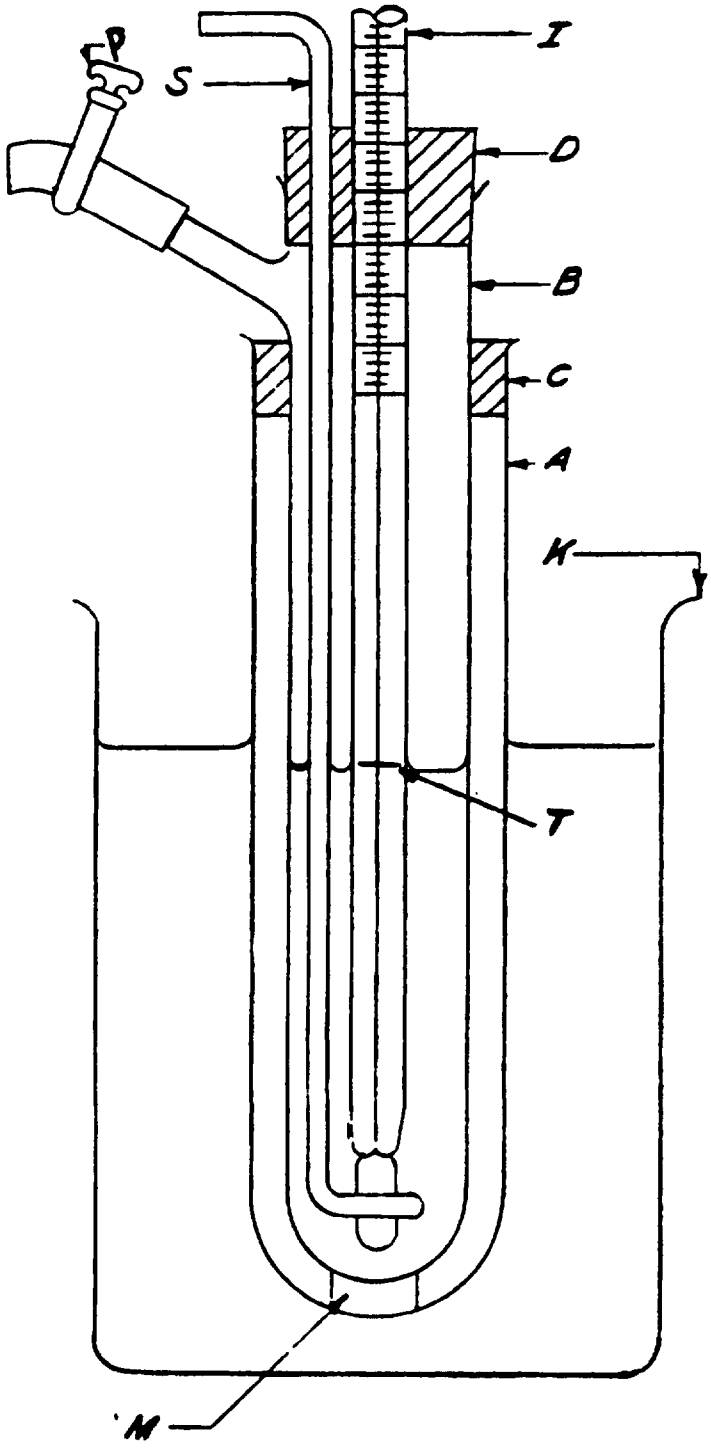


FIGURE 1.—*Apparatus for vacuum distillation mustard.*

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A- TEST TUBE 1 1/2" DIA. X 6 1/4" LONG.

B- TEST TUBE 1 3/8" DIA. X 7 1/4" LONG.

C- GOOCH TUBING CONNECTION.

D- STOPPER.

K- BEAKER

I- THERMOMETER GRADUATED TO READ IN TENTHS OF A DEGREE AND ARRANGED TO BE READ ABOVE STOPPER D.

M- CORK SPACER.

P- SCREW CLAMP.

S- STIRRER.

T- THERMOMETER IMMERSION MARK.

Figure 2.—Apparatus, freezing point.

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